

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question
For the linear function h , the graph of $y = h(x)$ in the xy -plane passes through the points $(7, 21)$ and $(9, 25)$. Which equation defines h ?

- Answer**
- A. $h(x) = \frac{1}{2}x - \frac{7}{2}$
- B. $h(x) = 2x + 7$
- C. $h(x) = 7x + 21$
- D. $h(x) = 9x + 25$

Correct Answer: B

Rationale
Choice B is correct. It’s given that the graph of the linear function h , where $y = h(x)$, passes through the points $(7, 21)$ and $(9, 25)$ in the xy -plane. An equation defining h can be written in the form $y = mx + b$, where $y = h(x)$, m represents the slope of the graph in the xy -plane, and b represents the y -coordinate of the y -intercept of the graph. The slope can be found using any two points, (x_1, y_1) and (x_2, y_2) , and the formula $m = \frac{(y_2 - y_1)}{(x_2 - x_1)}$. Substituting $(7, 21)$ and $(9, 25)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the slope formula yields $m = \frac{25 - 21}{9 - 7}$, which is equivalent to $m = \frac{4}{2}$, or $m = 2$. Substituting 2 for m and $(7, 21)$ for (x, y) in the equation $y = mx + b$ yields $21 = (2)(7) + b$, or $21 = 14 + b$. Subtracting 14 from each side of this equation yields $7 = b$. Substituting 2 for m and 7 for b in the equation $y = mx + b$ yields $y = 2x + 7$. Since $y = h(x)$, it follows that the equation that defines h is $h(x) = 2x + 7$.

- Choice A is incorrect. For this function, the graph of $y = h(x)$ in the xy -plane would pass through $(7, 0)$, not $(7, 21)$, and $(9, 1)$, not $(9, 25)$.
- Choice C is incorrect. For this function, the graph of $y = h(x)$ in the xy -plane would pass through $(7, 70)$, not $(7, 21)$, and $(9, 84)$, not $(9, 25)$.
- Choice D is incorrect. For this function, the graph of $y = h(x)$ in the xy -plane would pass through $(7, 88)$, not $(7, 21)$, and $(9, 106)$, not $(9, 25)$.